Ultimate Technologies Inc. — Take-Home Challenge

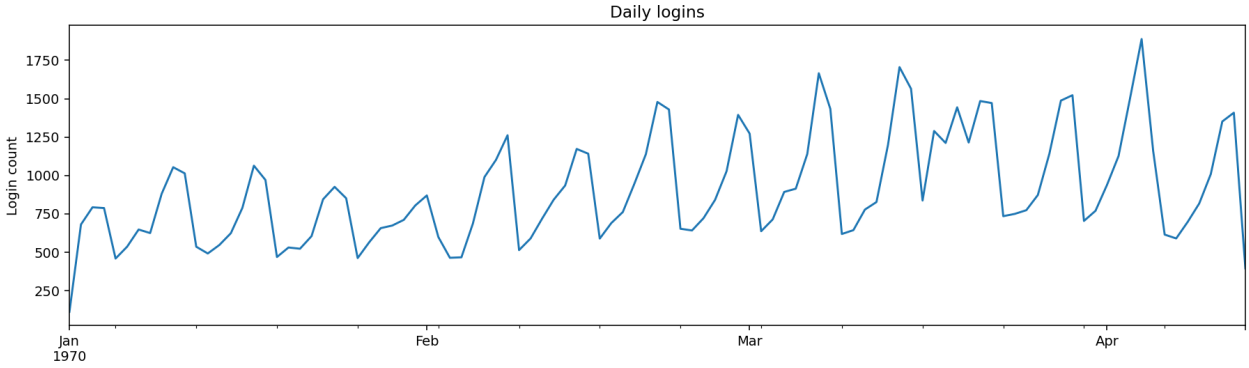
Completed analysis | Parts 1–3 | Based on the provided simulated data

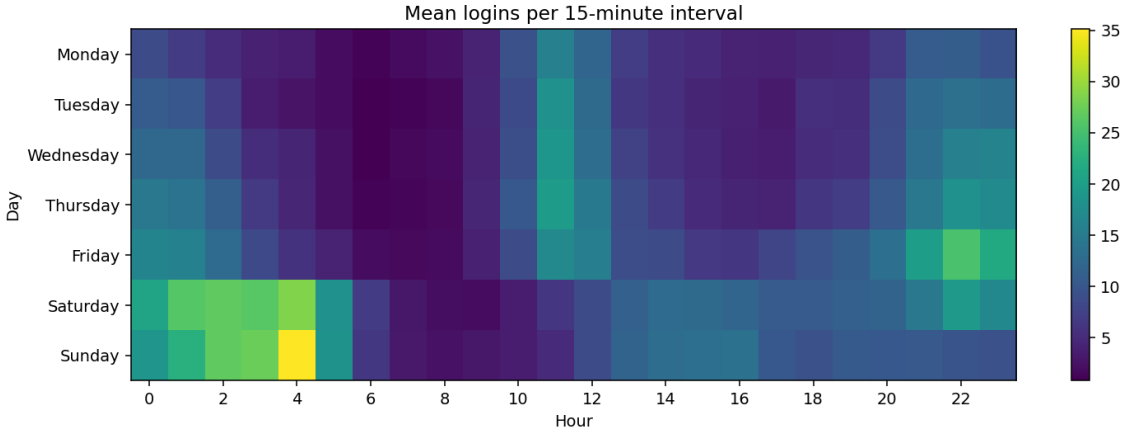
Executive summary

Analyzed 93,142 login events and 50,000 rider records. Observed 30-day retention was 37.61% (18,804 riders). Random Forest achieved held-out ROC-AUC 0.842, compared with 0.753 for logistic regression.

Part 1 — Exploratory analysis

Login timestamps span 1970-01-01 20:12:16 to 1970-04-13 18:57:38. All parsed successfully. We aggregated to contiguous 15-minute intervals, including 407 zero-count intervals. Saturday and Sunday average 13.46 and 12.62 logins per 15-minute bin, compared with Monday 6.21. Peaks cluster near 22:00–02:00, with a secondary midday peak. Repeated exact timestamps (877) were retained because they may be legitimate; timezone is unspecified and edge days are partial.





Part 2 — Experiment and metrics design

Primary success metric: percentage of eligible randomized driver partners completing at least one revenue trip in each city in a pre-specified 7-day window. This directly measures cross-city service and avoids selecting only drivers who crossed the bridge.

Implementation: Randomize eligible drivers 1:1, stratified by home city and baseline cross-city activity. Reimburse treatment-group tolls automatically, keep controls on existing policy, and run for four weeks with an initial ramp-up. Track trips, city coverage, online hours, reimbursement expense, cancellations, wait times, earnings, and contribution margin. Power the test in advance for an agreed minimum detectable absolute lift.

Statistical inference: Use a two-sided difference-in-proportions test at 5% significance, a 95% confidence interval, and a pre-specified baseline-adjusted logistic regression. Report intention-to-treat effect, absolute/relative lift, and cost per incremental cross-city driver. Analyze driver-level units, not correlated trips. Marketplace spillovers may violate individual randomization assumptions; if substantial, use a clustered or switchback design with appropriate inference.

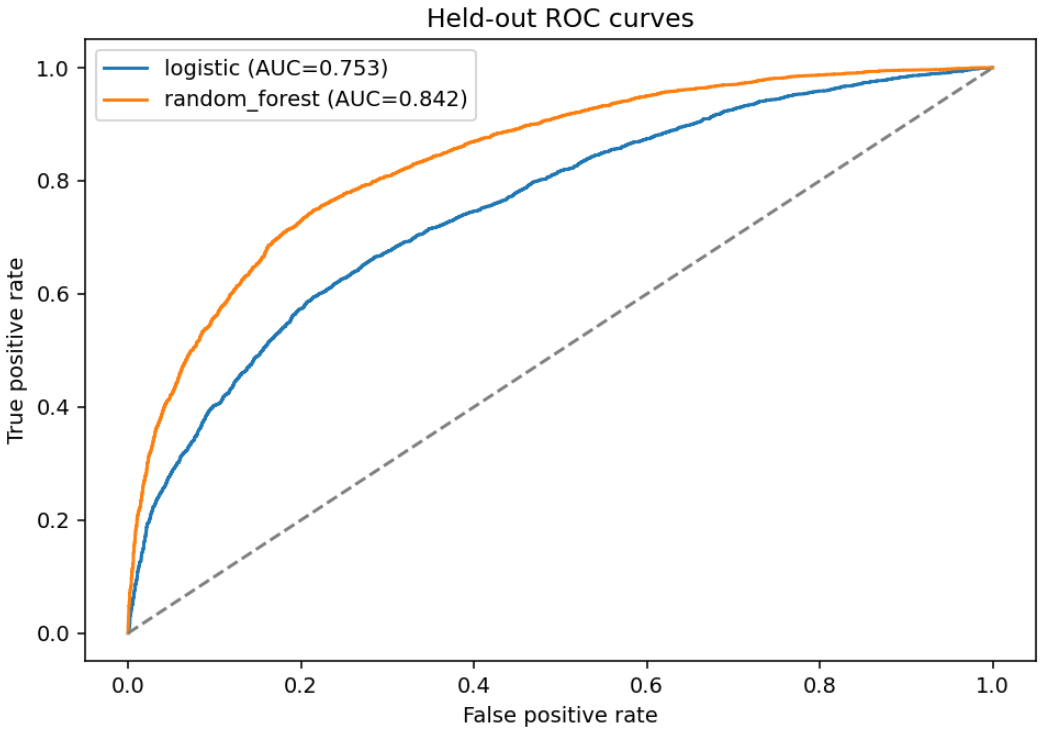
Recommendation rule: Roll out only if a statistically and operationally meaningful increase is supported and rider experience and incremental margin remain acceptable. A nonsignificant result is inconclusive if the experiment lacks power. Check seasonality, toll fraud, cross-city displacement, and differential treatment uptake.

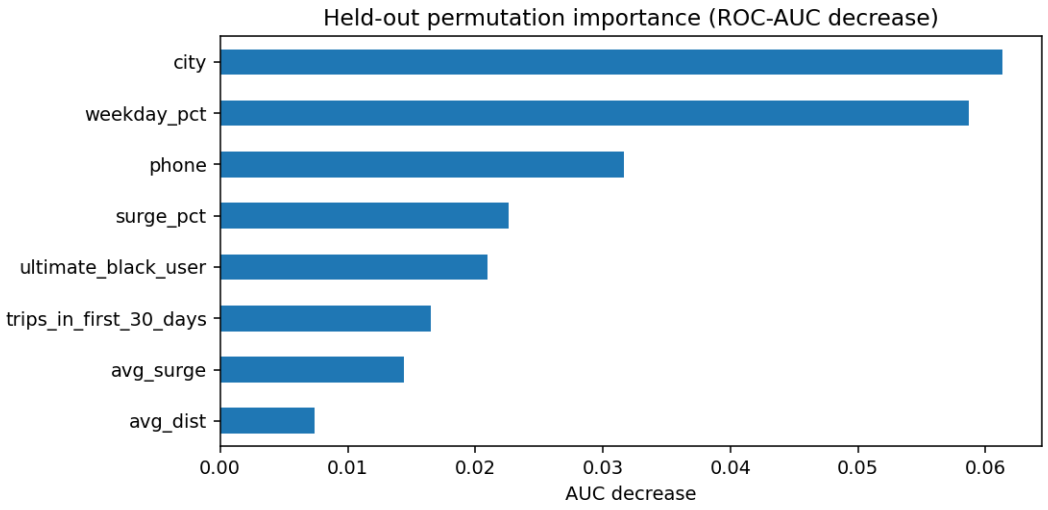
Part 3 — Predictive modeling

From 50,000 records, 18,804 (37.61%) had a last trip within 30 days of the latest observed trip date (2014-07-01); cutoff: 2014-06-01. Missing ratings: 8,122 rider-to-driver and 201 driver-to-rider; phone missing for 396. Eight identical rows were not removed because unique rider identifiers are unavailable.

We used city, phone, average trip distance, surge percentage, average surge, first-month trips, Ultimate Black use, and weekday trip percentage. Dates and overall-trip ratings were excluded to reduce leakage. We compared logistic regression and Random Forest on an 80/20 stratified split; preprocessing and imputation were fit only on the training data.

Model	ROC-AUC	Accuracy	Precision	Recall	F1
Logistic	0.753	0.715	0.662	0.493	0.565
Random Forest	0.842	0.776	0.741	0.621	0.676





Observed retention varies by city: Astapor 25.6%, King's Landing 62.8%, Winterfell 35.2%. Strong model signals include city, weekday percentage, phone and surge percentage. These are predictive associations rather than proven causal drivers.

Operational recommendations and limitations

Run randomized lifecycle and onboarding interventions for at-risk riders, evaluate city-level service reliability, and prioritize interventions by incremental retained riders and contribution margin. Validate on future signup cohorts and use strictly point-in-time feature definitions before deployment. The supplied feature documentation does not guarantee every trip aggregate is available at day 30; therefore this is an exploratory benchmark, not a deployment-ready forecast. The 30-day activity proxy also does not establish long-term retention.

Full executable code, metrics, and all figures are included in the companion repository files.